

Functional genetic report · longevity

Iron and Metals

Sérgio F · Report date: 02-10-2026

Iron (HFE, TMPRSS6, TF, TFR2), copper (ATP7B), zinc (SLC30A8, metallothioneins), selenium, magnesium and heavy metals handling. Only genes (chip + local imputation + polygenic scores). Plan target: ferritin < 100.

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Summary

46

SNPs analyzed

41

in the chip

4

imputed ($DR2 \geq 0.8$)

1

SNP not found in the data

3

Polygenic scores

Iron is essential, but in excess is one of the most direct accelerators of aging: it feeds the reaction of Fenton (hydroxy radicals), accumulates in lysosomes and senescent cells, favors ferroptosis and neuroinflammation. Therefore, the plan fixed ferritin below 100. Copper, zinc, selenium and magnesium are cofactors of antioxidant and mitochondrial enzymes; heavy metals compete with them.

In summary: genetics pulls iron upwards. Sérgio F carries HFE H63D (without C282Y) and the ferritin score of 1.08 million variants is at the 80th percentile. There is no sign of classical hemochromatosis, Wilson's disease or metallothionein deficiency. Genetic magnesium tends to the low side (3 loci).

Favourable

- **HFE C282Y absent** (rs1800562 GG): without classical hemochromatosis; HJV G320V absent.
- **Transferrin:** TF rs3811647 GG; ARNTL and FADS2 without the high iron alleles.
- **Copper:** ATP7B H1069Q absent (without Wilson carrier in the most common variant).
- **Zinc and metals:** SLC39A8 A391T absent; MT2A –5A/G without the G allele (normal expression of metallothionein).

Unfavourable

- **HFE H63D heterozygous** (rs1799945 CG): slightly increases iron, saturation and ferritin.
- **Ferritin score at the 80th percentile** (PGS003426, 1.08 M variants) and iron metabolism diseases in 76.
- **TF rs1799852 CT** (higher saturation allele, infrequent); TMPRSS6 A736V and TFR2 heterozygotes (typical).
- **Magnesium:** MUC1 rs4072037 CC, DCDC5 rs3925584 CC and ATP2B1 rs7965584 AG (lower magnesium alleles); score at the 29th percentile.
- **Not visible:** GSTM1/GSTT1 deletions (metal sequestration), ferroportin Q248H and rare variants of ATP7B, CP and TFR2.

Reading for ferritin < 100: with H63D and a ferritin score at the 80th percentile, Sérgio F tends to genetically accumulate iron above average. It is not hemochromatosis, but it means that the target of ferritin < 100 requires active management (without supplementary iron, absorption inhibitors at meals and, if the doctor agrees, blood donation).

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 41 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 2 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

The ferritin score (Bayesian method, 1.08 million variants) is the most informative. The simple iron scores made from few leading variants gave unstable results and are not used. The magnesium score is simple (21 loci of GWAS Catalog); a lower percentile indicates a tendency for lower magnesium.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
Ferritin (1.08 M variants, Bayesian method)	PGS003426	1 051 781 of 1 082 414 (97%)	p80	Typical / no judgement
Iron metabolism disorders (654 variants)	PGS001823	632 of 654 (97%)	p76	Typical / no judgement
Serum magnesium (GWAS Catalog score, 21 loci)	C_MAGNESIO	20 of 21 (95%)	p29	Typical / no judgement

HFE and hemochromatosis

HFE is the main gene of hereditary hemochromatosis. C282Y/C282Y causes the classical form; H63D has a small effect, but it is added to other factors (polygenic score, diet, fatty liver).

Polygenic scores of this theme: Iron metabolism disorders (654 variants) (PGS001823): the 76th percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
HFE C282Y	rs1800562	GG	chip	A	0	0.04	Main cause of hemochromatosis: C282Y/C282Y gives iron overload; 1 copy slightly rises the transferrin saturation. (strong evidence)	Favourable
HFE H63D	rs1799945	CG	chip	G	1	0.18	H63D: slightly increases serum iron, saturation and ferritin; alone rarely causes overload. (moderate evidence)	Unfavourable
HFE S65C	rs1800730	AA	chip	T	0	0.02	S65C: slight effect (relevant only with C282Y). (weak evidence)	Typical / no judgement

Favourable

- C282Y absent (rs1800562 GG).
- S65C absent (rs1800730 AA).

Unfavourable

- H63D heterozygous (rs1799945 CG).
- Iron metabolism disorders: score at the 76th percentile (typical, high side).

Interpretation: H63D carrier, without C282Y: the risk of hemochromatosis is low, but iron tends to be slightly higher than the average. It is the genetic basis of the ferritin rule < 100 of the plan.

Hepcidin, transferrin and other iron genes

Hepcidin is the hormone that stops iron absorption. TMPRSS6 (matriptase-2) turns off hepcidin; TFR2, HJV and HFE bind it. Transferrin (TF) transports iron in the blood.

Polygenic scores of this theme: Ferritin (1.08 M variants, Bayesian method) (PGS003426): the 80th percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
TMPRSS6 A736V	rs855791	AG	chip	G	1	0.61	Matriptase-2 (locking hepcidin): the V736 allele increases hepcidin and lower iron and saturation; A736 absorbs more iron. (moderate evidence)	Typical / no judgement
TMPRSS6	rs4820268	GA	chip	A	1	0.57	TMPRSS6: serum iron and saturation. (moderate evidence)	Typical / no judgement
TF	rs3811647	GG	chip	A	0	0.35	Transferrin: allele that increases transferrin (and low saturation). (moderate evidence)	Favourable
TF	rs1799852	CT	chip	T	1	0.13	Transferrin (L247L): transferrin and saturation. (weak evidence)	Unfavourable
TFR2	rs7385804	CA	chip	A	1	0.63	2 transferrin receptor (iron sensor in the liver): serum iron. (weak evidence)	Typical / no judgement
PCSK7	rs236918	GG	chip	C	0	0.12	PCSK7: transferrin saturation. (weak evidence)	Typical / no judgement
NAT2	rs4921915	AA	chip	A	2	0.75	NAT2 region: ferritin. (weak evidence)	Typical / no judgement
TEX14	rs411988	GA	chip	G	1	0.45	TEX14: ferritin.	Typical / no judgement
ARNTL	rs6486121	TT	chip	C	0	0.34	ARNTL (BMAL1, circadian clock): serum iron. (weak evidence)	Favourable
FADS2	rs174577	CC	chip	A	0	0.36	FADS2: serum iron (GWAS association). (weak evidence)	Favourable
SLC40A1 Q248H	rs11568350		SNP not found in the data	A	—	—	Ferroportin Q248H: higher ferritin (rare in the reference population). (moderate evidence)	SNP not found in the data
HFE2/HJV G320V	rs74315323	CC	chip	A	0	—	Hemojuvelin G320V: juvenile hemochromatosis (rare). (strong evidence)	Favourable

Favourable

- TF rs3811647 GG, ARNTL rs6486121 TT and FADS2 rs174577 CC: without the high iron alleles.
- HJV G320V absent.

Unfavourable

- Ferritin score at the 80th percentile.
- TF rs1799852 CT (infrequent allele of higher saturation).
- TMPRSS6 rs855791 AG and rs4820268 GA, TFR2 CA: heterozygotes (typical).

Interpretation: The polygenic base of ferritin is at the upper limit of the typical. TMPRSS6 A736V heterozygous: 1 allele that absorbs more iron and 1 that absorbs less (neutral effect). Ferroportin Q248H not found in the data (rare in the reference population).

Copper (ATP7B, ceruloplasmin)

Copper is cofactor of cytochrome c oxidase (mitochondria), SOD1, ceruloplasmin (which helps to export iron) and DAO (histamine). ATP7B eliminates excess copper by bile (Wilson's disease when failure).

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
ATP7B H1069Q	rs76151636	GG	chip	A	0	—	ATP7B H1069Q: the most common mutation of Wilson's disease (copper accumulation) in the reference population; carrier ~1%. (strong evidence)	Favourable
ATP7B K832R	rs1061472	TC	chip	—	—	—	ATP7B K832R: common polymorphism, possible mild effect on free copper. (weak evidence)	Typical / no judgement
ATP7B R952K	rs732774	CT	chip	—	—	—	ATP7B R952K: common polymorphism. (weak evidence)	Typical / no judgement
SMIM1	rs1175550	AA	chip	—	—	—	Serum copper locus (GWAS). (weak evidence)	Typical / no judgement
SELENBP1	rs2769264	TT	chip	—	—	—	Serum copper locus (GWAS). (weak evidence)	Typical / no judgement
CP ceruloplasmin	rs11708215	AG	imputation DR2=1.00	—	—	—	Ceruloplasmin region (copper transport, ferroxidase). (weak evidence)	Typical / no judgement

Favourable

- ATP7B rs76151636 GG

Unfavourable

- None.

Interpretation: Without H1069Q (the most common Wilson variant). K832R and R952K are common polymorphisms without clear clinical effect. Serum copper loci (SMIM1, SELENBP1) have no reliable effect direction. Nothing suggests accumulation or genetic deficit of copper.

Zinc and metallothioneins

Zinc: cofactor of >300 enzymes, immunity (thym, T cells) and insulin (ZnT8/SLC30A8 in beta cells). Metallothioneins store zinc and sequester cadmium, mercury and copper.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
SLC30A8 R325W (ZnT8)	rs13266634	CC	chip	C	2	0.72	Zinc transporter of beta cells (ZnT8): the R325 allele (C) increases the risk of 2 type diabetes. (moderate evidence)	Typical / no judgement
SLC39A8 A391T (ZIP8)	rs13107325	CC	chip	T	0	0.08	ZIP8 A391T: less manganese in the blood, altered glycosylation; it is associated with several traits (schizophrenia, Crohn, lipids). (moderate evidence)	Typical / no judgement
PPCDC	rs2120019	CC	chip	—	—	—	Zinc locus (GWAS). (weak evidence)	Typical / no judgement
CA1	rs1532423	AG	chip	—	—	—	Carbonic anhydrase 1 (zinc enzyme): erythrocyte zinc. (weak evidence)	Typical / no judgement
MT2A -5A/G	rs28366003	AA	imputation DR2=1.00	G	0	0.04	Metallothionein 2A: G allele reduces MT2A expression (less zinc reserve and less metal sequestration). (weak evidence)	Typical / no judgement
MT1A K51R	rs11640851	CC	chip	—	—	—	Metallothionein 1A: common polymorphism. (weak evidence)	Typical / no judgement

Favourable

- None.

Unfavourable

- None.

Interpretation: SLC30A8 rs13266634 CC: risk allele of diabetes type 2 in double dose, but is the most common genotype (typical). ZIP8 A391T absent and MT2A with normal expression: favourable for metal sequestration. There are no reliable markers of serum zinc in the chip.

Selenium and selenoproteins

Selenium enters selenoproteins (glutathione peroxidases, thioredoxin reductases, thyroid desiodases, selenoprotein P).

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
GPX1 Pro198Leu	rs1050450	GA	chip	A	1	0.34	Glutathione peroxidase 1 (selenoenzyme): Leu198 (A in strand +) has lower activity and responds more to selenium. (weak evidence)	Typical / no judgement
SELENOP Ala234Thr	rs3877899	CT	chip	—	—	—	Selenoprotein P (selenium transport): alters selenium distribution. (weak evidence)	Typical / no judgement
SELENOP	rs7579	CT	chip	—	—	—	Selenoprotein P (3'UTR). (weak evidence)	Typical / no judgement
SELENOF SEP15 1125G>A	rs5845	GG	chip	—	—	—	Selenoprotein F (15 kDa; folding of glycoproteins in the reticulum). (weak evidence)	Typical / no judgement
DMGDH	rs921943	CT	chip	—	—	—	Selenium locus in blood and nails (GWAS). (weak evidence)	Typical / no judgement
BHMT	rs6859667	TT	chip	T	2	0.96	Selenium locus (via methionine/ homocysteine). (weak evidence)	Typical / no judgement

Favourable

- None.

Unfavourable

- None.

Interpretation: GPX1 Pro198Leu heterozygous (typical): intermediate activity, responds well to the selenium of food. SELENOP and SELENOF with common polymorphisms. There is no sign of genetic deficit; excess selenium (supplements >200 µg) is associated with diabetes, so food arrives.

Magnesium

Magnesium: ATP cofactor, DNA stability and hundreds of enzymes; the deficit is associated with insulin resistance, inflammation and arrhythmias.

Polygenic scores of this theme: Serum magnesium (GWAS Catalog score, 21 loci) (C_MAGNESIO): the 29th percentile Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
TRPM6	rs3750425	CC	chip	—	—	—	Intestinal and renal magnesium channel TRPM6. (weak evidence)	Typical / no judgement
MUC1/ TRIM46	rs4072037	CC	chip	C	2	0.45	Locus MUC1: serum magnesium. (moderate evidence)	Unfavourable
SHROOM3	rs13146355	GA	chip	G	1	0.57	SHROOM3: serum magnesium (and renal function). (weak evidence)	Typical / no judgement
CNNM2	rs3740393	GG	chip	—	—	—	CNNM2 (magnesium transport): serum magnesium. (weak evidence)	Typical / no judgement
DCDC5	rs3925584	CC	chip	C	2	0.49	DCDC5: serum magnesium. (weak evidence)	Unfavourable
ATP2B1	rs7965584	AG	chip	G	1	0.16	Calcium pump ATP2B1: serum magnesium. (weak evidence)	Unfavourable

Favourable

- SHROOM3 GA (typical).

Unfavourable

- MUC1 rs4072037 CC and DCDC5 rs3925584 CC (low magnesium allele in double dose).
- ATP2B1 rs7965584 AG (infrequent allele).
- Magnesium score at the 29th percentile (typical, low side).

Interpretation: Genetic trend for magnesium slightly below average. TRPM6 and CNNM2 without defined effect direction in GWAS Catalog. Dapagliflozin tends to slightly increase serum magnesium, which plays in favor.

Heavy metals: absorption, sequestration and excretion

Absorption (DMT1), sequestration (metallothioneins, glutathione) and excretion (MRP2, AS3MT for arsenic) of lead, cadmium, mercury, arsenic and manganese. It overlaps with the theme of II phase detoxification (GST).

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
ALAD K59N (ALAD2)	rs1800435	CC	chip	—	—	—	ALAD2 binds more lead in red blood cells (lead in the highest blood, possibly less toxic in tissues). (weak evidence)	Typical / no judgement
GSTP1 Ile105Val	rs1695	AG	chip	G	1	0.34	GSTP1 Val105 (G): conjugation different from some metals and electrophiles; it is associated with more mercury / lead in some studies. (weak evidence)	Typical / no judgement
GSTO1 A140D	rs4925	CA	chip	—	—	—	GST omega 1 (arsenic biotransformation). (weak evidence)	Typical / no judgement
AS3MT	rs11191439	TC	chip	—	—	—	Arsenic methyltransferase: arsenic methylation efficiency. (weak evidence)	Typical / no judgement
SLC30A10	rs1776029	GG	imputation DR2=0.99	—	—	—	SLC30A10 manganese exporter: manganese in blood. (weak evidence)	Typical / no judgement
SLC11A2 DMT1	rs224589	TG	imputation DR2=0.99	—	—	—	DMT1: intestinal absorption of iron and divalent metals (cadmium, lead, manganese). (weak evidence)	Typical / no judgement
ABCC2 MRP2	rs717620	CC	chip	—	—	—	MRP2: biliary excretion of glutathione conjugates with metals (mercury). (weak evidence)	Typical / no judgement

Favourable

- None.

Unfavourable

- None.

Interpretation: No major effect changes were observed. GSTP1 Ile105Val heterozygous (typical). The GSTM1/GSTT1 deletions, which weigh on the elimination of metals and are very common, are not visible in the chip (the null profile indicates probable GSTM1 by a marker SNP). ALAD, GSTO1, AS3MT, SLC30A10, DMT1 and MRP2: common polymorphisms without reliable effect direction.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Oxidative stress / Redox imbalance	HFE H63D; ferritin p80; TF rs1799852	C282Y absent; MT2A normal	The free iron feeds the Fenton reaction: it is the most direct link of this report to aging.
Lysosome dysfunction	H63D + ferritin p80 (lysosomal iron)	—	Iron accumulates in lysosomes with age and reduces autophagy.
Cell Senescence	Trend for high iron	—	Senescent cells accumulate iron and ferritin.
Release of Mitochondrial DAMPs	Trend for high iron	—	The excess iron damages the mitochondria.
Oxidative Phosphorylation Decline	—	Normal ATP7B (cytochrome c oxidase copper)	Neutral.
Neuroinflammation	Iron (accumulates in the ganglia of the base with age)	Normal ATP7B; SLC30A10 without defined risk	Mild weakness.
Insulin Resistance / Hyperglycemia and Intolerance to Glucose	SLC30A8 CC (common genotype); high iron	—	High iron aggravates insulin resistance.
Mitochondrial Calcium Deregulation	Low genetic magnesium (MUC1, DCDC5, ATP2B1)	—	Magnesium stabilizes ATP and calcium channels.
Immunosenescence	—	Normal metallothionein; normal ZIP8	Neutral.
DNA Damage and DNA Repair Deficit	GSTM1/GSTT1 not visible	—	Heavy metals damage DNA; visible genetic handling is typical.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
Iron: no supplements	No iron or multivitamins with iron; fortified cereals and avoided viscera, red meat with moderation (as in the plan). suggestion (not applied)	HFE H63D; ferritin p80; target < 100
Iron: absorption at meals	Tea, breakfast or dairy at meals and calcium at meals (as in the plan); supplemental vitamin C always ≥ 2 h away from food. More integral fiber and legumes (phytates), always: the more fiber, the better. suggestion (not applied)	H63D; TF rs1799852
Iron: blood donation	The most effective and inexpensive way to maintain ferritin < 100 with this genetic is the donation of regular blood, if the doctor agrees. suggestion (not applied)	H63D + ferritin p80 score
Iron: natural chelators	Curcumin and quercetin chelate iron, but pulsating them while ALT is high is a suggestion (not applied; the real stack did not change); the monthly senolytic pulse (fisetin + quercetin) only with normal ALT. suggestion (not applied)	V2 plan rules
Magnesium	Magnesium by food (vegetables, legumes, seeds, cocoa) and, if necessary, 200–400 mg/day of glycinate or threonate at the end of the day. Avoid chronic proton pump inhibitors (low magnesium). suggestion (not applied)	MUC1 CC, DCDC5 CC, ATP2B1 AG; p29 score
Zinc and copper	If you take zinc, ≤ 25 mg/day and with 1–2 mg of copper (the isolated zinc lowers the copper, which DAO and ceruloplasmin need). suggestion (not applied)	Normal ATP7B; DAO (profile)
Selenium	For food: 1–2 Brazil nuts per week, fish and eggs. No selenium supplements > 100–200 µg. suggestion (not applied)	GPX1 heterozygous
Heavy metals	Purified fish oil (already taken) instead of large predators (spadarte, large tuna) frequently; rice well washed and cooked in a lot of water (arsenic); filter the water if the channeling is old. suggestion (not applied)	GSTM1/GSTT1 not visible; typical handling
Vitamin D and omega-3	Maintain D3 10 000 IU and omega-3 3.2–4 g/day, as in the plan. suggestion (not applied)	Coordination with the plan
What not to do	Do not use telomerase activators. Without iron, without iron, without vitamin C with meals. suggestion (not applied)	V2 plan rules

Limitations

- The chip does not read the GSTM1/GSTT1 deletions (very common and relevant to metals); the profile indicates probable null GSTM1 by a marker SNP, without confirmation.
- Rare variants of hemochromatosis (TFR2, HJV, HAMP, SLC40A1) and Wilson (centenarians of variants in ATP7B) are not covered, except those tested here.
- There are no reliable polygenic scores for copper, zinc and selenium; these themes are based on isolated SNPs of small effect.
- The percentiles compare with the reference population (n=633). Real iron depends heavily on the diet, blood loss and liver.
- Only genes: does not use blood analysis.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (statistical imputation, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Frequency of alleles calculated locally in the reference population (n=633).
- Effect alleles: cured from literature or GWAS Catalog (local copy of 13-09-2026), with the verified sense (iron, ferritin and higher saturation = unfavourable; lower magnesium = unfavourable).
- Scores: PGS003426 (ferritin, Toivonen 2023), PGS001823 (iron metabolism diseases, Privé 2022) and a simple magnesium score (21 loci of GWAS Catalog). Percentages vs. reference population (n=633) with exactly the same variants.
- Main references: Feder 1996 (HFE, Nat Genet); Benyamin 2014 and Bell 2021 (iron markers, Nat Common / Common Biol); Finberg 2008 (TMPRSS6, Nat Genet); Ferenci 2003 (ATP7B H1069Q); Sladek 2007 (SLC30A8, Nature); Evans 2013 (copper and zinc, Hum Mol Genet); Cornelis 2015 (selenium, Hum Mol Genet); Meyer 2010 (magnesium, PLoS Genet); Stockwell 2017 (ferroptosis, Cell).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).